

TC_LCD_001

Case Description

LCD_Init should initialize without errors using the configuration from LCD_CFG.h

Case Pre-Conditions

DIO module initialized

Priority

↑ High

```

5 // 4-bit configuratio
6 #define LCD_RS      F
7 #define LCD_EN      F
8 #define LCD_D4      F
9 #define LCD_D5      F
10 #define LCD_D6      F
11 #define LCD_D7      F

```

- ✓ Verify that the static function **LCD_Instruction** is behaving correctly with DIO modules function **DIO_SetPin**  2 

Comment

LCD_Instruction function is verified completely. In the attachments provided is one of the interactions between the two modules data transferred correctly as required.

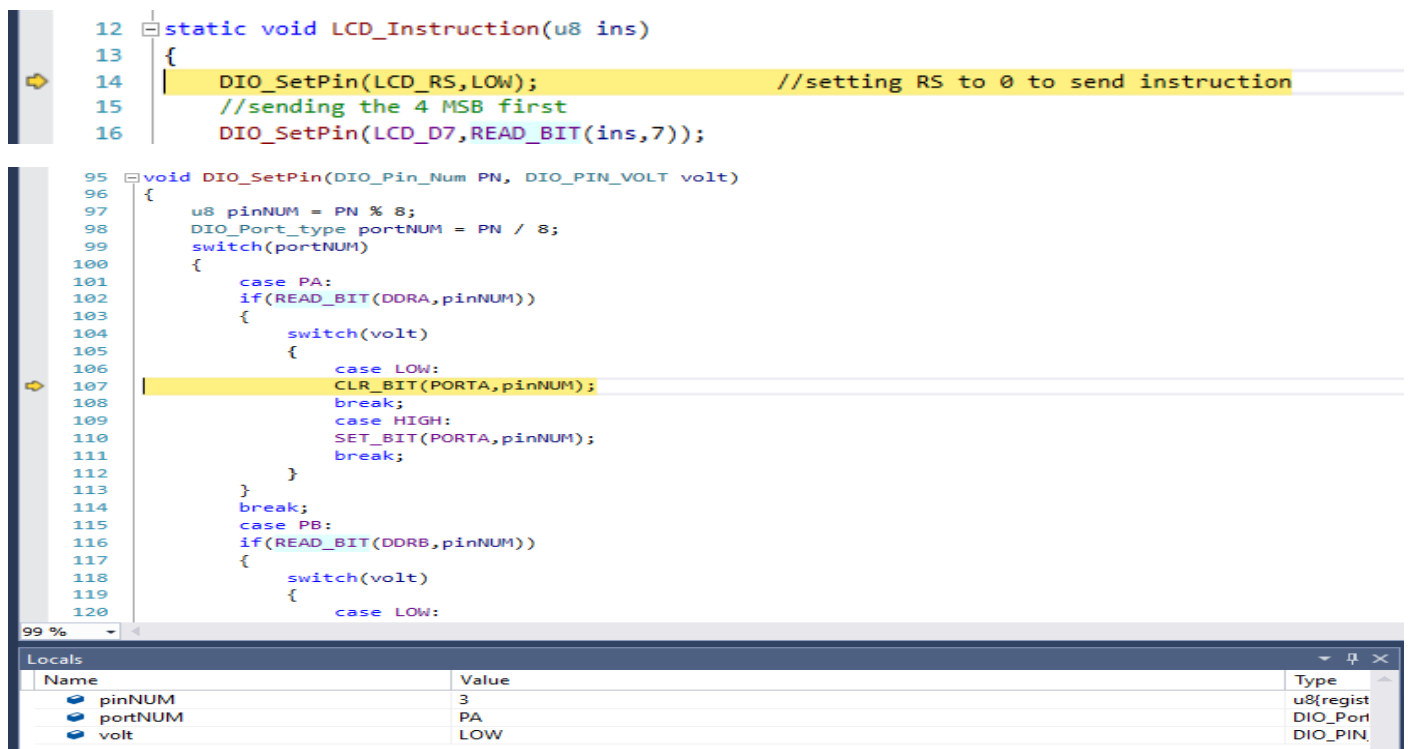
Attachments

 lcd_ins%201.png

 lcd_ins%202.png

- ✓ Verify that all five (5) instructions are executed correctly and LCD initialized in Proteus successfully

Run attachments



```

12 static void LCD_Instruction(u8 ins)
13 {
14     DIO_SetPin(LCD_RS,LOW);           //setting RS to 0 to send instruction
15     //sending the 4 MSB first
16     DIO_SetPin(LCD_D7,READ_BIT(ins,7));

```

```

95 void DIO_SetPin(DIO_Pin_Num PN, DIO_PIN_VOLT volt)
96 {
97     u8 pinNUM = PN % 8;
98     DIO_Port_type portNUM = PN / 8;
99     switch(portNUM)
100     {
101         case PA:
102             if(READ_BIT(DDRA,pinNUM))
103             {
104                 switch(volt)
105                 {
106                     case LOW:
107                         CLR_BIT(PORTA,pinNUM);
108                         break;
109                     case HIGH:
110                         SET_BIT(PORTA,pinNUM);
111                         break;
112                 }
113             }
114             break;
115         case PB:
116             if(READ_BIT(DDRB,pinNUM))
117             {
118                 switch(volt)
119                 {
120                     case LOW:

```

Name	Value	Type
pinNUM	3	u8{regist
portNUM	PA	DIO_Port
volt	LOW	DIO_PIN

TC_LCD_002

[Execution](#) [Run History](#) [Retries](#)

Case Description

Verifying LCD_WriteChar function. Character 'A' should be displayed at the cursor position (default start position: x=2, y=1)

Case Pre-Conditions

- DIO and LCD modules initialized
- Run Proteus simulation loaded with the project hex file

Priority

↑ High

```

16 //////////////////////////////////////////////////
17 int main()
18 {
19     DIO_Init();
20     LCD_Init();
21     while(1)
22     {
23         LCD_WriteChar('A');
24     }
25 }
26 //////////////////////////////////////////////////

```

- ✓ Adjust Proteus single step time to 100us to be able to catch one instruction **only** of the LCD_WriteChar function

Attachments

proteus%20step%20time.png

- ✓ Verify that 'A' char is shown **one** time at the default starting position x=2 & y=1

Attachments

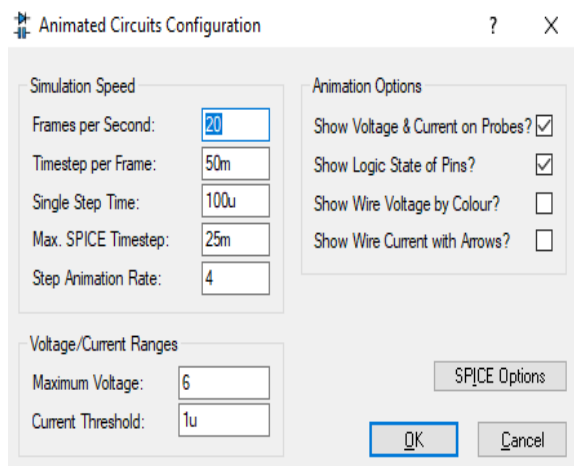
proteus%20writechar.png

- ✓ Verify that the interaction between LCD and DIO modules behaves correctly

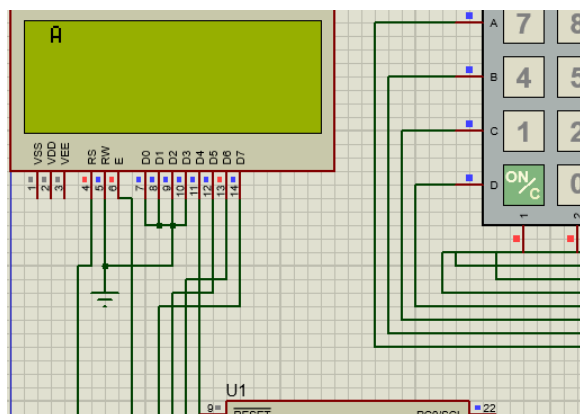
Comment

LCD_WriteChar interacts with DIO_SetPin sucessfully

Run attachments



TC_LCD_002 Proteus step time



TC_LCD_002 WriteChar on LCD

Case Description

Case Pre-Conditions

- ### Priority

```

17 //
18 int main()
19 {
20     GPIO_Init();
21     LCD_Init();
22     while(1)
23     {
24         LCD_WriteChar('R');
25         delay_ms(100);
26         LCD_Clear();
27     }
28 }

```

- ## Attachments

 lcd_clr2.png

- ## Attachments

 lcd_clr3.png

Run attachments

The screenshot shows the 'Animated Circuits Configuration' dialog box in the top left corner. It has two main sections: 'Simulation Speed' and 'Animation Options'.

Simulation Speed:

- Frames per Second: 20
- Timestep per Frame: 50m
- Single Step Time: 5m
- Max. SPICE Timestep: 25m
- Step Animation Rate: 4

Animation Options:

- Show Voltage & Current on Probes? ☒
- Show Logic State of Pins? ☒
- Show Wire Voltage by Colour? ☐
- Show Wire Current with Arrows? ☐

At the bottom of the dialog are 'Voltage/Current Ranges' (Maximum Voltage: 6, Current Threshold: 1u), 'SPICE Options' (disabled), and 'OK' and 'Cancel' buttons.

The background shows a circuit simulation interface. It features a green rectangular area at the top, likely a display for the circuit's state. Below this is a circuit diagram with various components and pins labeled. The pins are numbered 1 through 14, with labels VSS, VDD, VEE, RS, RW, E, D0, D1, D2, D3, D4, D5, D6, D7. A 'RESET' button is visible at the bottom right. The status bar at the bottom indicates '5 Message(s)' and 'PAUSED: 0.076000250s'.

TC LCD 003 LCD after 15 ms

TC_LCD_004

[Execution](#) [Run History](#) [Retries](#)

Case Description

Test writing string on the screen using **LCD_WriteString** with **Hello World** looped with **LCD_CLR()**.

Case Pre-Conditions

- DIO and LCD initialized
- TC_LCD_003 run to be successful. As we are gonna use **LCD_CLR()**
- Run Proteus simulation

Priority

↑ High

```

17 int main()
18 {
19     DIO_Init();
20     LCD_Init();
21     while(1)
22     {
23         LCD_WriteString("Hello
24         LCD_CLR();
25     }
26 }

```

✓ Generate .hex file from the main function as shown in the attached image to be executed in Proteus simulation.

✓ Observe the expected output in Proteus simulation. 

 1 

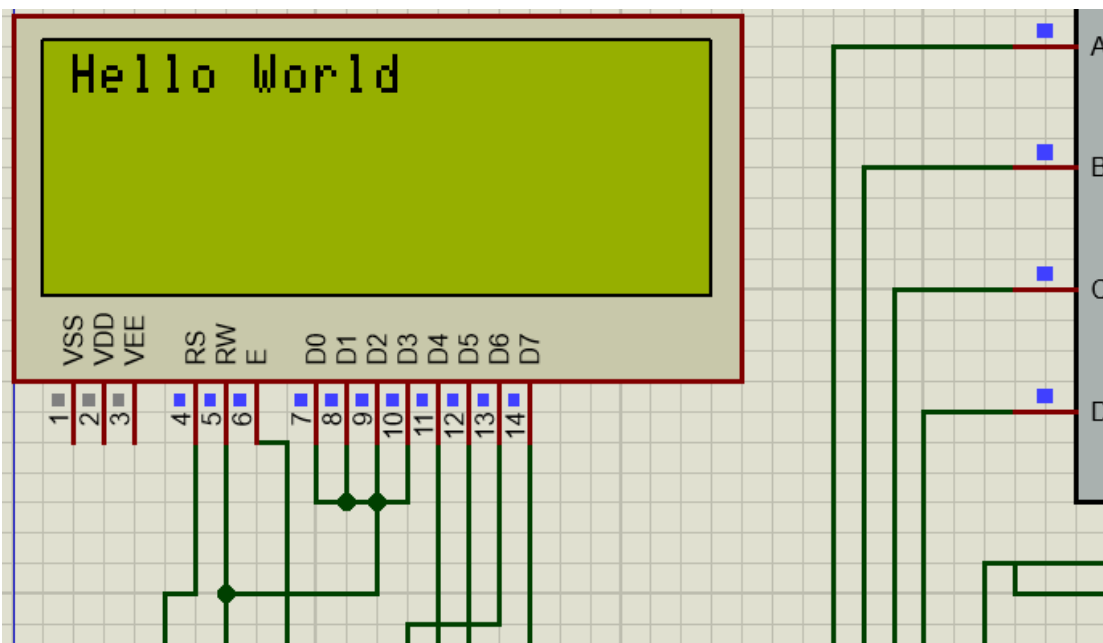
Expected result

Hello World is blinking in Proteus Simulation

Attachments

 lcd_writestr2.png

Run attachments



TC_LCD_005 (1st run)

Case Description

Test LCD_GoTo function using LCD_WriteChar to confirm the position of the writing.

Case Pre-Conditions

- DIO and LCD initialized
- TC_LCD_003 run to be successful. As we are gonna use LCD_WriteChar & LCD_CLR()
- Run Proteus simulation

Priority

High



Comment

There are 4 cells lost in row 3 and 4, the cursor starts from column 5 in each row.

Attachments

Proteus%20failed.m
p4

- Generate .hex file from the main function as shown in the attached image to be executed in Proteus simulation.
- Adjust Proteus animation options to clearly see the transitions of the char in every cell in the LCD.

Attachments

proteus%20animation%20options.png

show the expected output as mp4 or gif

Expected result

char T is written and cleared in every cell (4×16)

Defect report

← LCD few cells not working (TC_LCD_005)

Edit Delete

Actual Result

LCD rows (3&4) columns (0 to 3) not working, cursor starts from column 5

Attachments

Proteus failed.mp4

Test Run Results

CASE	AUTHOR	RUN	TITLE	TIME	STATUS
CECP-11	Ahmed Magdy	R-13	TC_LCD_005	7m 22s	Failed

Comments

Ahmed Magdy QA
Defect in LCD_GoTo function resolved by adjusting the values of the cursor placement position



5 days ago

Status

Resolved

Severity

Critical

Reporter

Ahmed Magdy

Assignee

Ahmed Magdy

Created

27 Jun 2024 02:40:37

```
99
100 void LCD_GoTo(u8 line, u8 cell)
101 {
102     if (line==1)
103     {
104         LCD_Instruction(0x80+cell);
105     }
106     else if (line==2)
107     {
108         LCD_Instruction(0x80+0x40+cell);    // 0xC0: Sets cursor to line 2 of display + cell
109     }
110     else if (line==3)
111     {
112         LCD_Instruction(0x80+0x10+cell);    // 0x90: Sets cursor to line 3 of display + cell
113     }
114     else if (line==4)
115     {
116         LCD_Instruction(0x80+0x50+cell);    // 0xD0: Sets cursor to line 4 of display + cell
117     }
118 }
119
```

TC_LCD_005 Defect resolved

Video link: [Proteus failure](#)

TC_LCD_005 (2nd run)

[Execution](#) [Run History](#) [Retries](#)

Case Description

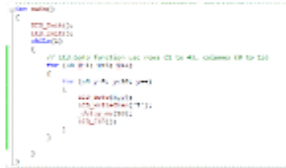
Test LCD_GoTo function using LCD_WriteChar to confirm the position of the writing.

Case Pre-Conditions

- DIO and LCD initialized
- TC_LCD_003 run to be successful. As we are gonna use LCD_WriteChar & LCD_CLR()
- Run Proteus simulation

Priority

↑ High



- ✓ Generate .hex file from the main function as shown in the attached image to be executed in Proteus simulation.

- ✓ Adjust Proteus animation options to clearly see the transitions of the char in every cell in the LCD. ✓ 

Attachments

 proteus%20animation%20options.png

- ✓ show the expected output as mp4 or gif ✓ 

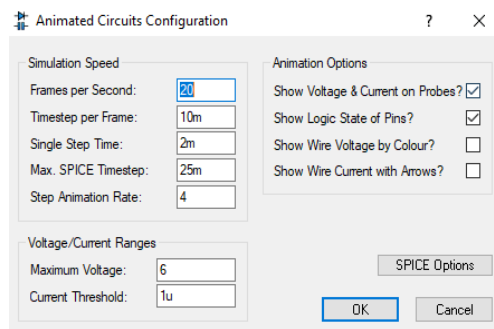
Expected result

char T is written and cleared in every cell (4×16)

Attachments

 proteus%20defect%20resolved.mp4

Run attachments



TC_LCD_005 Animation Options

Video link: [Proteus after defect resolved](#)

TC_LCD_006

[Execution](#) [Run History](#) [Retries](#)

Case Description

Test writing string on the screen using `LCD_WriteNumber`.

Case Pre-Conditions

- DIO and LCD initialized
- Run Proteus simulation

Priority

↑ High

```

17 int main()
18 {
19     LCD_Init();
20     LCD_WriteString("0123456789012345");
21     while(1)
22     {
23         LCD_WriteNumber(0);
24         for (i=0; i<16; i++)
25             if (i%16 == 0)
26             {
27                 LCD_WriteString("0123456789012345");
28                 delay(1000);
29             }
30     }
31 }

```

- ✓ Generate .hex file from the main function as shown in the attached image to be executed in Proteus simulation.

- ✓ show the expected output as mp4 or gif ▼

1

Input data

loop numbers from 0 to 9 in 4×16 cells

Expected result

{0123456789012345}
 {0123456789012345}
 {0123456789012345}
 {0123456789012345}

to be printed out on the 4×16 LCD

Attachments

proteus%20writenum.mp4

Video link: [Proteus WriteNum](#)